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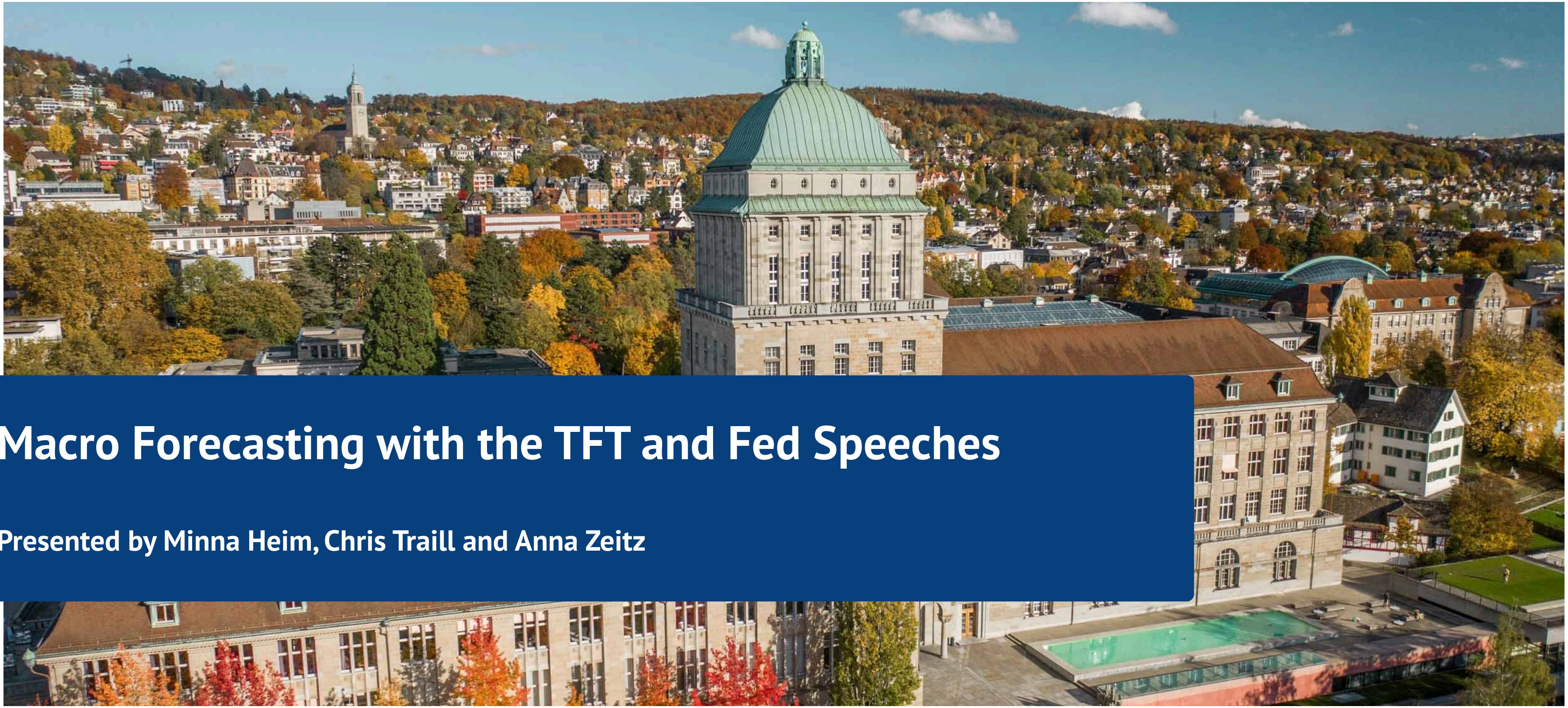

Macro Forecasting with the TFT and Fed Speeches

Presented by Minna Heim, Chris Traill and Anna Zeitz

Motivation

- Macro forecasts matter for
  - Governments
  - Financial markets
  - **Central banks**

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$\Rightarrow$ Can we do better than standard macro models?

Research Question

- The Fed communicates with the public through official statements and speeches
- It may communicate additional information:
  1. **Superior information**: private data and internal models not available to the public
  2. **Forward guidance**: signals about future policy to guide market expectations

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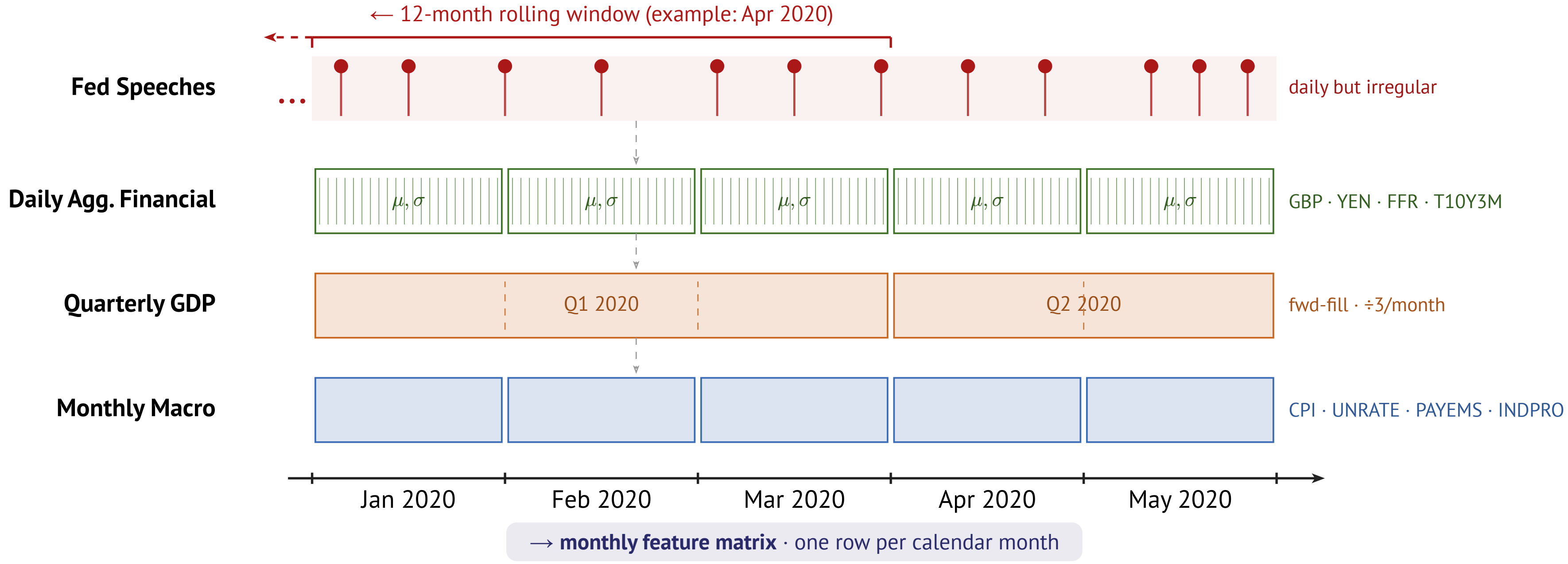
$\Rightarrow$ **Do Fed speeches contain information useful to forecast macroeconomic indicators?**

Data Alignment



Models

Benchmarks

$$\text{AR(p)} \qquad y_t = c + \sum_{i=1}^{p} \varphi_i y_{t-i} + \varepsilon_t$$

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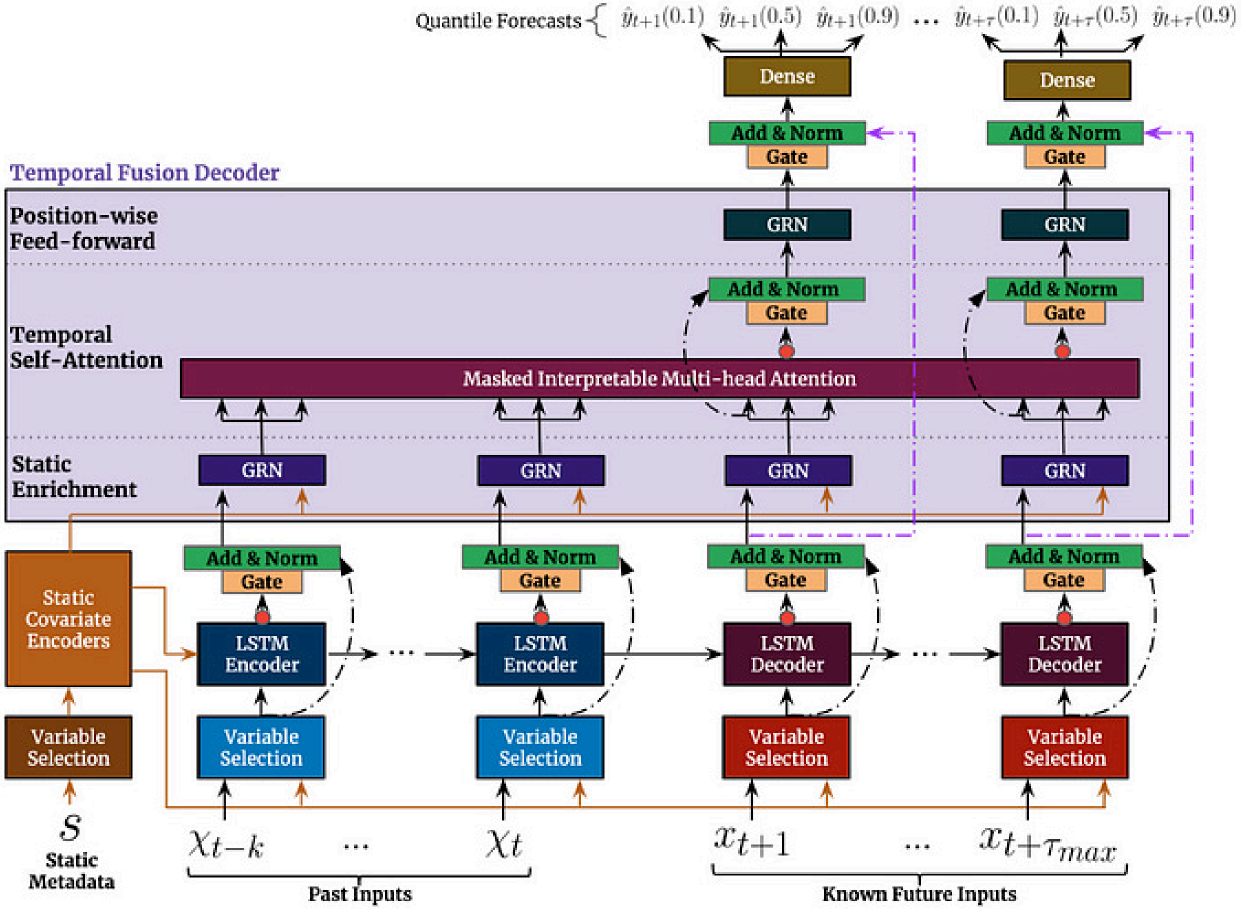
$$\text{AR(p)} \qquad y_t = c + \sum_{i=1}^{p} \varphi_i y_{t-i} + \varepsilon_t$$

$$\text{ARIMA(p, d, q)} \qquad \Delta^d y_t = c + \sum_{i=1}^{p} \varphi_i \, \Delta^d y_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t$$

- Integration: d-th difference of y_t
- MA component: q lags of residuals ε_t

Temporal Fusion Transformers



Speech Embeddings

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Two models chosen for complementary perspectives:

1. **FinBERT**: BERT-based, pre-trained on financial statements (Yang et al., 2020)
2. **FOMC-RoBERTa**: fine-tuned on FOMC data, hawkish/dovish classification (Shah et al., 2023)

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Processing: chunk-and-average (512-token windows, 128-token overlap)

- Assumption: information spread across entire speech, not just the beginning

Dimensionality reduction: 768 $\Rightarrow X$ dims via Principal Component Analysis or Factor Analysis

- X is treated as hyperparameter

Speech Alignment

Speeches are held at **irregular** and **daily** frequency: how to align with monthly horizon?

1. **Mean aggregation**: unweighted mean over rolling window of X months
2. **Exponential decay**: $w_t = \exp(-\lambda \cdot d_t)$ where d_t = days before month m
   - More recent speeches receive higher weight
   - λ treated as fixed; speech window X tuned as hyperparameter
3. **Attention-based**: key = embedding (after dimensionality reduction), query = mean macro state
   - Given current macro conditions, which past speeches are most informative?
   - Fitted on training data only $\Rightarrow$ no leakage!

All methods also track: voter speech ratio, chair speech ratio, gender share

Hyperparameter Tuning

Tune in two stages:

1. **Stage 1: Architecture (Macro-Only TFT)**
   - Bayesian search via Optuna (TPE sampler)
   - 2-fold CV, up to 50 trials
   - Includes: encoder length, hidden size, normalizer
2. **Stage 2: Speech Embedding Params**
   - Architecture fixed from Stage 1
   - Tune: aggregation, reduction, PCA dims, speech window

Evaluation Protocol

- **Walk-forward cross-validation**: expanding window, never use future data
- **Metric**: MAE (mean absolute error), RMSE ????

ADJUST HERE

- **Multi-step forecasting**: predictions in non-overlapping 12-month windows
  - Context: training data + model's own previous predictions
  - Same assumption for AR, ARIMA and TFT $\Rightarrow$ fair comparison

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$\Rightarrow$ All models evaluated on the same information set

Results

Results

- Incl. Tables, Figures

Robustness Checks

- Shuffling Embeddings
- Using unrelated text (German Texts of Franz Kafka)
- Use not full speech but only first 512 tokens of speech???

Future Work

- Hyperparameter tuning: switch to one stage tuning! Also: context-dependent attention
- Try out SSMs for forecasting
  - Combine with TFT: replace LSTM with SSM for sequence encoder
- Add Vintages: use real-time macro data (as available at forecast time) for more realistic evaluation
- Extend dataset: Working Paper (Lustenberger et al., 2026)
  - 10,000+ speeches since 1914, continuously updated

Bibliography

Lustenberger, T., Rossi, E., & Zeitz, A. (2026). *Central Bank Communication: New Data and Stylized Facts From a Century of Fed Speeches* (Working Paper No. 6/2026). https://doi.org/tbd

Shah, A., Paturi, S., & Chava, S. (2023,). Trillion Dollar Words: A New Financial Dataset, Task & Market Analysis. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4447632

Yang, Y., UY, M. C. S., & Huang, A. (2020,). *FinBERT: A Pretrained Language Model for Financial Communications*. https://arxiv.org/abs/2006.08097

Appendix

Variable List: Macroeconomic Variables

1. **Target variables**
   - CPI, UNRATE, GDP (quarterly: forward-fill missing months) $\Rightarrow$ AR and ARIMA only take targets
2. **Covariates**
   - Monthly variables: payroll (PAYEMS), industrial production (INDPRO), hours worked in manufacturing (AWHMAN), OECD composite leading indicator (USACLI)
   - Weekly / daily variables: exchange rates (GBP, YEN), effective federal funds rate (FFR), Fed balance sheet (WALCL)
     - Higher-frequency variables: take mean **and** std per month (volatility)
   - Lags at 1, 2, 6, 12 months for all target variables
3. **Preprocessing**: log-difference of CPI, GDP, PAYEMS, INDPRO, AWHMAN

Data Exploration:

- show head of final dataframe (train,e.g.)

Data Exploration 2:

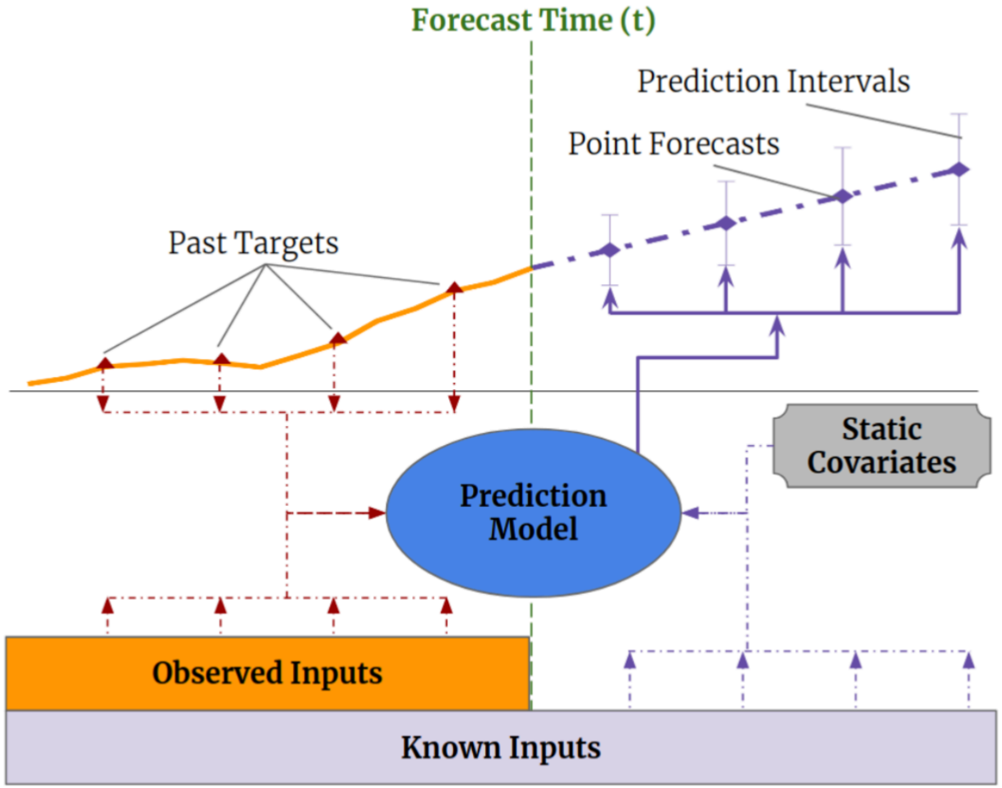
- show plotted data (in dfb)

Variable List

The TFT takes four different types of variables:

1. **Static categoricals**: time-invariant, categorical
   - Identifier and frequency of each time series
2. **Static reals**: time-invariant, continuous
   - Number downloads time series from FRED, how long time series has existed
3. **Time-Varying Known Reals**: time-varying, known
   - Day of week, month, year; if day is holiday
   - FOMC Meeting dates
4. **Time-Varying Unknown Reals**: time-varying, unknown
   - Macroeconomic variables, speech embeddings, FOMC dissents

TFT: Input Types



Dimensionality Reduction

FinBERT and FOMC-RoBERTa embeddings are of dimension 768 per speech

$\Rightarrow$ Dimensionality reduction needed for TFT input

We test two methods (number of components X is hyperparameter, fit only on training data):

1. **PCA**: projects embeddings onto X principal components
   - 20 components $\approx$ 80% of variance explained
2. **Factor Analysis (FA)**: models embeddings as linear combination of X latent factors
   - Unlike PCA: explicitly models noise, focuses on **shared** variance
   - Parameters estimated via EM algorithm

Based on your tuning code, here are the two appendix slides filled out: typst

Hyperparameter Tuning: Stage 1 Search Space

Architecture tuned on macro-only TFT (per target × horizon):

Parameter	**Type**	**Range / Values**
Encoder length	int	12 – 48
Hidden size	categorical	$\{8, 16, 32, 64, 128, 256\}$
Hidden continuous size	categorical	$\{2, 4, 8, 16\}$
Dropout	float	0.05 – 0.55
Learning rate	log-uniform	10^{-4} – 0.15
Normalizer	categorical	{encoder-none, group}

Fixed: `lstm_layers = 2`, `attention_head_size = 2`, `max_epochs = 50`, `patience = 15`

Hyperparameter Tuning: Stage 2 Search Space

Speech embedding params tuned (architecture fixed from Stage 1):

Parameter	Type	Range / Values
Aggregation	categorical	{mean, decay, attention}
Reduction	categorical	{pca, fa}
PCA components (X)	int	5 – 30
Speech window (months)	int	3 – 12

Tuned separately for each of 9 combinations: {CPI, GDP, UNRATE} $\times$ {3, 6, 12} months

Hyperparameter Tuning: Stage 2 – Optimal Hyperparameters

	FOMC-RoBERTa			**FinBERT**		
Target	**h=3**	**h=6**	**h=12**	**h=3**	**h=6**	**h=12**
CPI	Mean PCA, n = 21 W = 3 MAE = 0.001847	Attention FA, n = 10 W = 8 MAE = 0.001849	**Attention** **PCA, n = 12** **W = 9** **MAE = 0.001842**	**Attention** **FA, n = 20** **W = 5** **MAE = 0.001846**	**Decay** **PCA, n = 23** **W = 3** **MAE = 0.001846**	Mean PCA, n = 8 W = 7 MAE = 0.001847
GDP	**Mean** **PCA, n = 8** **W = 7** **MAE = 0.00154**	Attention FA, n = 12 W = 8 MAE = 0.00171	Attention FA, n = 5 W = 10 MAE = 0.00152	Attention FA, n = 6 W = 12 MAE = 0.00170	**Attention** **FA, n = 10** **W = 8** **MAE = 0.00157**	**Decay** **PCA, n = 20** **W = 12** **MAE = 0.00151**
UNRATE	Decay FA, n = 12 W = 11 MAE = 1.31293	Attention FA, n = 6 W = 12 MAE = 1.31501	**Mean** **PCA, n = 8** **W = 7** **MAE = 1.11597**	**Mean** **FA, n = 23** **W = 5** **MAE = 1.26152**	**Mean** **PCA, n = 14** **W = 4** **MAE = 1.30896**	Mean PCA, n = 25 W = 4 MAE = 1.20808

where W refers to the speech window size and n to the optimal number of components for the dimensionality reduction. The best result per target $\times$ horizon is shown in bold.